

August 30, 1923.

Mr. W. deC. Ravenel,
Administrative Assistant,
U. S. National Museum.

Dear Sir:

I herewith submit my report on the expedition to the Dinosaur National Monument, near Jensen, Utah.

In accordance with your letter of instructions dated May 10, 1923, I left Washington on the evening of May 11, arriving at the Dinosaur National Monument late in the evening of May 14. The following morning, in company with Mr. Earl Douglass, the resident representative of the Carnegie Museum, I was hurriedly shown over the quarry, as he was leaving that morning for Craig, Colorado, to superintend the shipping of a consignment of fossil specimens to Pittsburgh.

In the quarry were two *Diplodocus* skeletons that had been partially uncovered by the Carnegie people, and after a preliminary survey, it was at once decided to first excavate the skeleton which was most complete (No. 355), and then secure such parts of the second (No. 340) as were needed to complete the first.

It should be mentioned that before his departure, Mr. Douglass, representing the Carnegie Museum, turned over to me four large blocks and several smaller ones containing portions of the neck of specimen No. 340. One block belonging to this specimen was shipped to Pittsburgh, but will, I understand, be turned over to the National Museum as soon as certain bones of another individual are extracted from it. Likewise, one of the blocks presented to us contains bones which we are expected to prepare and forward to the Carnegie Museum.

The week following my arrival at the quarry was spent in getting equipment together, employing necessary laborers, and perfecting other working arrangements. Regular work in the quarry was begun on May 24. I was fortunate in securing the services of three men who had had previous experience in the work under Mr. Douglass, and this employment of trained men contributed much to the successful outcome of the operations. The work was further facilitated by the arrival on June 5 of Mr. Norman H. Boss, whose skilful handling of fossil specimens was of the greatest assistance.

From May 24 the work proceeded continuously with but slight interruptions, such being mainly due to my offer to furnish Mr. Douglass with the necessary help in loading his 70 large boxes which were to be hauled to the railroad by truck - an offer which seemed fully justified in consideration of the generosity of the Carnegie Museum in turning over to us the several blocks of specimen No. 340, mention of which has already been made.

On June 30, following your instructions, the first consignment of material, consisting of 13 boxes having a total weight of 19,445 pounds, was loaded for shipment to Washington. After some delays on the road, due to break down of wagons, this consignment reached the railroad at Price, Utah, on July 13, and was shipped on the same date.

During my absence at Price, a cloudburst at the quarry carried away some of our lumber, nails, and tools, little of which was recovered. Some time was also lost due to this rain, in repairing roads, bailing out the quarry, etc., but otherwise the work proceeded regularly up to the close of operations on August 9.

Arrangements had previously been made for freighting the final shipment to Price, and on August 11, all boxes were loaded and on their way, arriving there in splendid condition on August 19. The date of arrival falling on Sunday, shipment could not be made until the following day, when the 21 boxes comprising the consignment were loaded directly into a car. This lot of material weighed 31,180 pounds. On the evening of August 20, Mr. Boss and I left Price, arriving in Washington on the evening of August 24.

I have advised you from time to time as to the progress of the work, but for the completion of this report, will briefly sum up the results.

Sufficient skeletal material of *Diplodocus* was obtained to enable us to mount a good skeleton of this extinct dinosaur. In size there is every reason for believing it will exceed 80 feet in length, with a height at the hips in excess of 14 feet. The only important parts of the skeleton unrepresented are the skull, axis, tip of the tail, and fore feet. We already have in the Museum two good skulls of this genus, of which one will be available for use in the mount. The tip of the tail, feet, and other parts missing can be supplied by casts if not original bones, which I am sure the Carnegie Museum will be willing to supply.

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In detail, so far as can be determined from the unprepared materials, the two skeletons have the following bones present:

Skeleton No. 355

Vertebrae (articulated)

2 posterior cervical vertebrae
10 dorsal vertebrae
4 sacral "
30+ caudal "
10 ribs of left side
4+ " " right "
Numerous chevron bones

Pelvis

2 ilia
2 ischia
2 pubes

Hind limb

left femur
" tibia
" fibula
hind foot

Fore limb

left scapula
" humerus
2 ulnae
2 radii

Skeleton No. 340.

Vertebrae (articulated)

13 cervical vertebrae
3 dorsal "

Fore limb

Right scapula
" humerus
" radius

Sternal plate.

In addition to the above enumerated material, a considerable amount of miscellaneous fossils representative of the Morrison fauna was collected. In all, the collection filled 34 large cases, having a total weight of 50,625 pounds or over 25 tons.

The total cost of the expedition, exclusive of freight charges from Price, Utah, to Washington, was \$3201.16. This total includes \$759.38 paid for freighting the boxes from the quarry to the shipping point, so that the actual cost for excavating is \$2441.78.

Very respectfully,

C. W. Gilmore

Curator, Vertebrate Paleontology.